Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Ulys Dairy

Physical address of dairy:

8118 Sikes RD	Dixon	Solano	95620
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1983

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

0111-X110-X070-XXXX

B. OPERATORS

Mendoza, Ulysses

Operator name: Mendoza, Ulysses

Telephone no.: Landline: ; Cellular: (209) 595-1201

8118 Sikes RD	Dixon	CA	95620
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Mendoza, Ulysses

Legal owner name: Mendoza, Ulysses

Telephone no.: Landline: ; Cellular: (209) 595-1201

8118 Sikes RD	Dixon	CA	95620
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	350	210	78	78
Number under roof	950	134	0	0	0	0
Maximum number	950	134	350	210	78	78
Average number	950	134	350	210	78	78
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 76 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 33,187.74 *tons per reporting period*

Total nitrogen from manure: 412,928.12 *lbs per reporting period*

After ammonia losses (30% loss applied): 289,049.68 *lbs per reporting period*

Total phosphorus from manure: 68,332.53 *lbs per reporting period*

Total potassium from manure: 185,678.67 *lbs per reporting period*

Total salt from manure: 478,120.80 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 45,906,000 *gallons*

Total nitrogen generated: 87,546.79 *lbs*

Total phosphorus generated: 8,838.69 *lbs*

Total potassium generated: 88,482.25 *lbs*

Total salt generated: 818,898.86 *lbs*

	45,906,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	45,906,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW G	Ground water
IW M	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
G1	75	75	2	process wastewater	0111-X110-X070-XXXX
G2	37	37	2	process wastewater	0111-X110-X070-XXXX
Miles 1	36	36	2	both	0111-X110-X050-XXXX
Miles 2	35	35	2	both	0111-X110-X050-XXXX
Miles 3	58	58	2	both	0111-X110-X050-XXXX
Miles 4	58	58	2	both	0111-X110-X050-XXXX
Totals for areas that were used for application	299	299	12		
Totals for areas that were not used for application					
Land application area totals	299	299	12		

B. CROPS AND HARVESTS

G1

Field name: G1

10/30/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 75 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	1,387.50 *ton*	As-is		69.8	4,140.00	830.00	4,370.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	18.50	153.18	30.71	161.69	1,005.66

G1

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	2,115.00 *ton*	As-is		69.6	4,350.00	840.00	4,660.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	245.34	47.38	262.82	1,320.21

G2

Field name: G2

10/31/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 37 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	621.60 *ton*	As-is		67.2	5,050.00	1,040.00	6,840.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	16.80	169.68	34.94	229.82	980.85

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,024.90 *ton*	As-is		68.8	4,340.00	670.00	4,090.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.70	240.44	37.12	226.59	1,088.94

Miles 1

Field name: Miles 1

11/02/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	622.00 *ton*	As-is		67.4	4,990.00	900.00	5,780.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	17.28	172.43	31.10	199.73	1,058.92

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	990.00 *ton*	As-is		68.6	4,400.00	700.00	4,270.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.50	242.00	38.50	234.85	1,139.82

Miles 2

Field name: Miles 2

11/02/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 35 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	568.60 *ton*	As-is		65.0	5,500.00	890.00	5,100.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	16.25	178.70	28.92	165.71	1,023.48

Miles 2

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	1,021.50 *ton*	As-is		70.8	3,940.00	430.00	4,190.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.19	229.98	25.10	244.58	1,516.96

Miles 3

Field name: Miles 3

11/01/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 58 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	953.50 *ton*	As-is		69.2	5,850.00	950.00	6,400.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	16.44	192.34	31.24	210.43	941.79

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,728.40 *ton*	As-is		72.0	3,720.00	570.00	4,100.00		7.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.80	221.71	33.97	244.36	1,214.89

Miles 4

Field name: Miles 4

11/01/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 58 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	923.40 *ton*	As-is		66.4	5,140.00	1,010.00	7,370.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	15.92	163.66	32.16	234.67	887.99

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,682.00 *ton*	As-is		68.8	3,930.00	650.00	4,010.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.00	227.94	37.70	232.58	1,393.39

NUTRIENT BUDGET

A. LAND APPLICATIONS

G1 - 10/30/2022: Wheat, silage, soft dough

Field name: G1

Crop: Wheat, silage, soft dough — Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.23	11.48	116.13	814.54	1,470,000.00 *gal*
Application event totals		84.23	11.48	116.13	814.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	125.62	3.60	73.88	434.61	1,260,000.00 *gal*
Application event totals		125.62	3.60	73.88	434.61	

G1 - 06/23/2023: Corn, silage

Field name: G1

Crop: Corn, silage — Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW G	Ground water	24.62	0.00	0.00	724.53	9,792,000.00 *gal*
Application event totals		24.62	0.00	0.00	724.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.86	6.62	54.21	643.03	3,360,000.00 *gal*
IW G	Ground water	14.48	0.00	0.00	426.20	5,760,000.00 *gal*
Application event totals		59.35	6.62	54.21	1,069.23	

G1 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.74	6.45	52.85	626.96	3,276,000.00 *gal*
IW G	Ground water	14.12	0.00	0.00	415.54	5,616,000.00 *gal*
Application event totals		57.86	6.45	52.85	1,042.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.18	6.37	52.18	618.92	3,234,000.00 *gal*
IW G	Ground water	13.94	0.00	0.00	410.21	5,544,000.00 *gal*
Application event totals		57.12	6.37	52.18	1,029.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW G	Ground water	23.54	0.00	0.00	692.57	9,360,000.00 *gal*
Application event totals		23.54	0.00	0.00	692.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.06	6.20	50.82	602.84	3,150,000.00 *gal*
IW G	Ground water	13.58	0.00	0.00	399.56	5,400,000.00 *gal*
Application event totals		55.64	6.20	50.82	1,002.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.62	6.29	51.50	610.88	3,192,000.00 *gal*
IW G	Ground water	0.01	0.00	0.00	0.40	5,472.00 *gal*
Application event totals		42.63	6.29	51.50	611.29	

G2 - 10/31/2022: Wheat, silage, soft dough

G2 - 10/31/2022: Wheat, silage, soft dough

Field name: G2

Crop: Wheat, silage, soft dough — Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.18	9.97	100.88	707.61	630,000.00 *gal*
Application event totals		73.18	9.97	100.88	707.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	152.78	4.38	89.86	528.58	756,000.00 *gal*
Application event totals		152.78	4.38	89.86	528.58	

G2 - 06/25/2023: Corn, silage

Field name: G2

Crop: Corn, silage — Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.47	6.71	54.94	651.72	1,680,000.00 *gal*
IW G	Ground water	14.68	0.00	0.00	431.96	2,880,000.00 *gal*
Application event totals		60.15	6.71	54.94	1,083.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.20	6.37	52.19	619.14	1,596,000.00 *gal*
IW G	Ground water	13.95	0.00	0.00	410.36	2,736,000.00 *gal*
Application event totals		57.14	6.37	52.19	1,029.49	

G2 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.06	6.20	50.82	602.84	1,554,000.00 *gal*
IW G	Ground water	13.58	0.00	0.00	399.56	2,664,000.00 *gal*
Application event totals		55.64	6.20	50.82	1,002.40	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW G	Ground water	23.49	0.00	0.00	691.13	4,608,000.00 *gal*
Application event totals		23.49	0.00	0.00	691.13	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.06	6.20	50.82	602.84	1,554,000.00 *gal*
IW G	Ground water	13.58	0.00	0.00	399.56	2,664,000.00 *gal*
Application event totals		55.64	6.20	50.82	1,002.40	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.79	5.87	48.07	570.26	1,470,000.00 *gal*
IW G	Ground water	12.84	0.00	0.00	377.96	2,520,000.00 *gal*
Application event totals		52.63	5.87	48.07	948.22	

10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW G	Ground water	21.29	0.00	0.00	626.34	4,176,000.00 *gal*
Application event totals		21.29	0.00	0.00	626.34	

Miles 1 - 11/02/2022: Wheat, silage, soft dough

Field name: Miles 1

Crop: Wheat, silage, soft dough

Plant date: 11/02/2022

Miles 1 - 11/02/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	233.89	57.35	73.05	0.00	1,080.00 *ton*
Application event totals		233.89	57.35	73.05	0.00	

Miles 1 - 06/27/2023: Corn, silage

Field name: Miles 1

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	125.02	25.54	34.14	1,205.87	600.00 *ton*
Application event totals		125.02	25.54	34.14	1,205.87	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	24.25	0.00	0.00	763.73	4,608,000.00 *gal*
Application event totals		24.25	0.00	0.00	763.73	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.56	6.72	55.06	653.08	1,638,000.00 *gal*
IW M	Ground water	14.78	0.00	0.00	465.40	2,808,000.00 *gal*
Application event totals		60.34	6.72	55.06	1,118.48	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.49	0.00	0.00	739.87	4,464,000.00 *gal*
Application event totals		23.49	0.00	0.00	739.87	

Miles 1 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.49	0.00	0.00	739.87	4,464,000.00 *gal*
Application event totals		23.49	0.00	0.00	739.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	22.73	0.00	0.00	716.00	4,320,000.00 *gal*
Application event totals		22.73	0.00	0.00	716.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	21.97	0.00	0.00	692.13	4,176,000.00 *gal*
Application event totals		21.97	0.00	0.00	692.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	21.22	0.00	0.00	668.27	4,032,000.00 *gal*
Application event totals		21.22	0.00	0.00	668.27	

Miles 2 - 11/02/2022: Wheat, silage, soft dough

Field name: Miles 2

Crop: Wheat, silage, soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	233.89	57.35	73.05	0.00	1,050.00 *ton*
Application event totals		233.89	57.35	73.05	0.00	

Miles 2 - 06/25/2023: Corn, silage

Field name: Miles 2

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.07	7.09	58.08	688.96	1,680,000.00 *gal*
IW M	Ground water	15.59	0.00	0.00	490.97	2,880,000.00 *gal*
Application event totals		63.65	7.09	58.08	1,179.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.66	6.74	55.18	654.52	1,596,000.00 *gal*
IW M	Ground water	14.81	0.00	0.00	466.42	2,736,000.00 *gal*
Application event totals		60.47	6.74	55.18	1,120.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.46	6.56	53.73	637.29	1,554,000.00 *gal*
IW M	Ground water	14.42	0.00	0.00	454.15	2,664,000.00 *gal*
Application event totals		58.88	6.56	53.73	1,091.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.38	0.00	0.00	736.46	4,320,000.00 *gal*
Application event totals		23.38	0.00	0.00	736.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	22.99	0.00	0.00	724.18	4,248,000.00 *gal*
Application event totals		22.99	0.00	0.00	724.18	

Miles 2 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Process wastewater	42.06	6.20	50.82	602.84	1,470,000.00 *gal*
IW M	Ground water	13.64	0.00	0.00	429.60	2,520,000.00 *gal*
Application event totals		55.70	6.20	50.82	1,032.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	21.43	0.00	0.00	675.09	3,960,000.00 *gal*
Application event totals		21.43	0.00	0.00	675.09	

Miles 3 - 11/01/2022: Wheat, silage, soft dough

Field name: Miles 3

Crop: Wheat, silage, soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.69	10.18	102.97	722.25	1,008,000.00 *gal*
Application event totals		74.69	10.18	102.97	722.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	151.61	4.35	89.17	524.53	1,176,000.00 *gal*
Application event totals		151.61	4.35	89.17	524.53	

Miles 3 - 06/19/2023: Corn, silage

Field name: Miles 3

Crop: Corn, silage Plant date: 06/19/2023

Miles 3 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	103.46	21.13	28.25	997.96	800.00 *ton*
Application event totals		103.46	21.13	28.25	997.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.13	6.95	56.95	675.60	2,730,000.00 *gal*
IW M	Ground water	15.29	0.00	0.00	481.45	4,680,000.00 *gal*
Application event totals		62.42	6.95	56.95	1,157.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.52	0.00	0.00	740.69	7,200,000.00 *gal*
Application event totals		23.52	0.00	0.00	740.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.05	0.00	0.00	725.88	7,056,000.00 *gal*
Application event totals		23.05	0.00	0.00	725.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	22.81	0.00	0.00	718.47	6,984,000.00 *gal*
Application event totals		22.81	0.00	0.00	718.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	22.58	0.00	0.00	711.06	6,912,000.00 *gal*
Application event totals		22.58	0.00	0.00	711.06	

Miles 3 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	22.10	0.00	0.00	696.25	6,768,000.00 *gal*
Application event totals		22.10	0.00	0.00	696.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	21.87	0.00	0.00	688.84	6,696,000.00 *gal*
Application event totals		21.87	0.00	0.00	688.84	

Miles 4 - 11/01/2022: Wheat, silage, soft dough

Field name: Miles 4

Crop: Wheat, silage, soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.47	9.33	94.39	662.06	924,000.00 *gal*
Application event totals		68.47	9.33	94.39	662.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	151.61	4.35	89.17	524.53	1,176,000.00 *gal*
Application event totals		151.61	4.35	89.17	524.53	

Miles 4 - 06/22/2023: Corn, silage

Field name: Miles 4

Crop: Corn, silage Plant date: 06/22/2023

Miles 4 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	103.46	21.13	28.25	997.96	800.00 *ton*
Application event totals		103.46	21.13	28.25	997.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.86	7.06	57.83	685.99	2,772,000.00 *gal*
IW M	Ground water	15.52	0.00	0.00	488.86	4,752,000.00 *gal*
Application event totals		63.38	7.06	57.83	1,174.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.52	0.00	0.00	740.69	7,200,000.00 *gal*
Application event totals		23.52	0.00	0.00	740.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.28	0.00	0.00	733.28	7,128,000.00 *gal*
Application event totals		23.28	0.00	0.00	733.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.05	0.00	0.00	725.88	7,056,000.00 *gal*
Application event totals		23.05	0.00	0.00	725.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	23.05	0.00	0.00	725.88	7,056,000.00 *gal*
Application event totals		23.05	0.00	0.00	725.88	

Miles 4 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	22.34	0.00	0.00	703.66	6,840,000.00 *gal*
Application event totals		22.34	0.00	0.00	703.66	

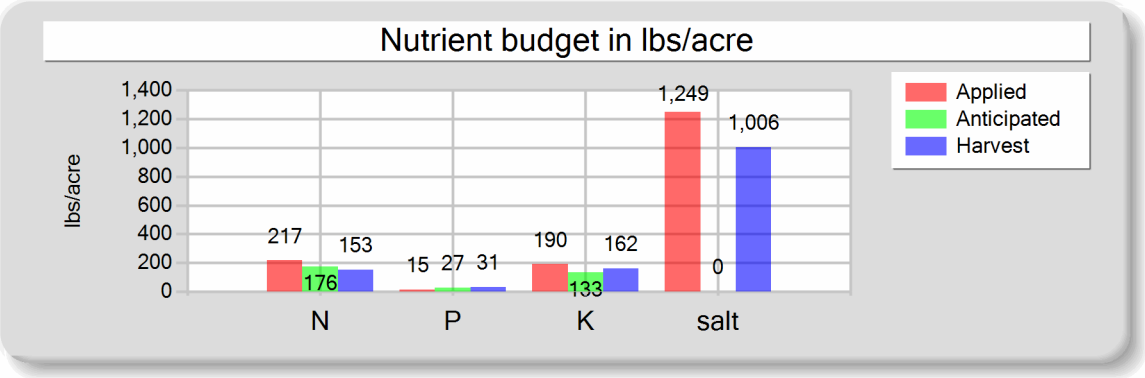
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW M	Ground water	21.87	0.00	0.00	688.84	6,696,000.00 *gal*
Application event totals		21.87	0.00	0.00	688.84	

B. NUTRIENT BUDGET

G1 - 10/30/2022: Wheat, silage, soft dough

Field name: G1 Crop: Wheat, silage, soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.85	15.09	190.01	1,249.15
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.85	15.09	190.01	1,249.15
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	153.18	30.71	161.69	1,005.66
Nutrient balance	63.67	-15.62	28.32	243.49
Applied to removed ratio	1.42	0.49	1.18	1.24

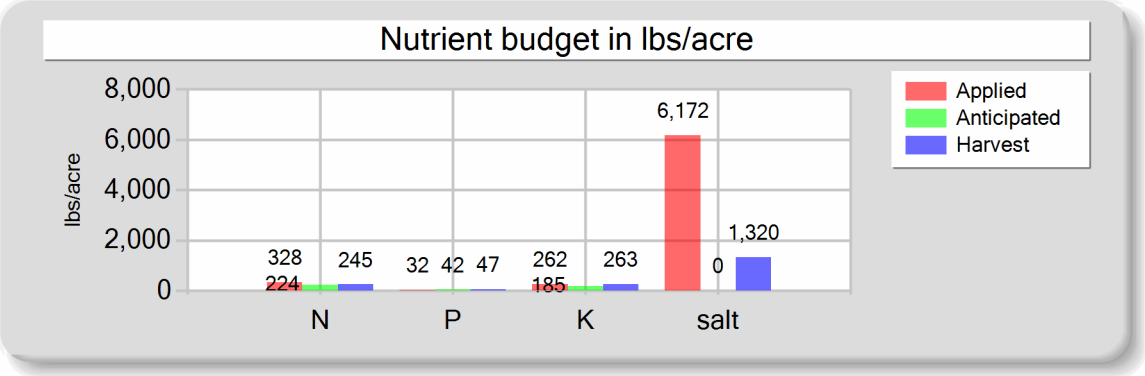
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,730,000.00	*gallons*
100.54	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

G1 - 06/23/2023: Corn, silage

Field name: G1 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.46	31.93	261.56	3,102.63
Fresh water	104.30	0.00	0.00	3,069.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.76	31.93	261.56	6,171.65
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	245.34	47.38	262.82	1,320.21
Nutrient balance	82.42	-15.45	-1.26	4,851.43
Applied to removed ratio	1.34	0.67	1.00	4.67

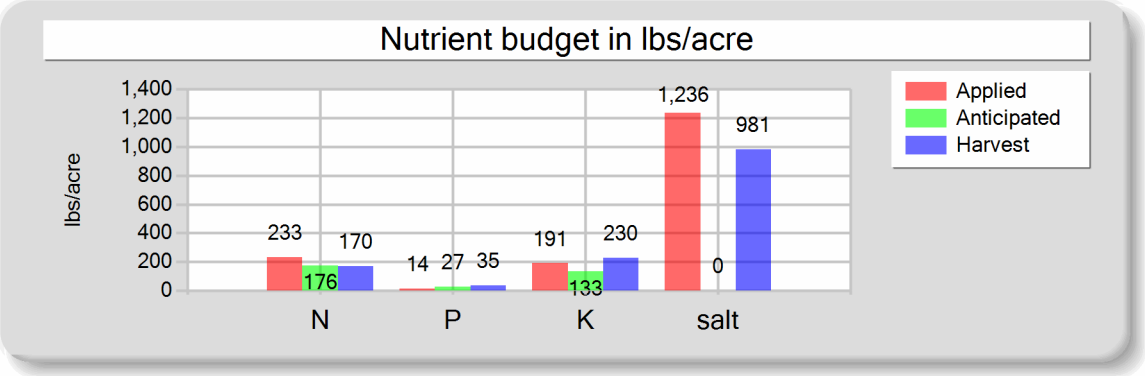
Fresh water applied	
41,477,472.00	*gallons*
1,527.47	*acre-inches*
20.37	*inches/acre*

Process wastewater applied	
16,212,000.00	*gallons*
597.03	*acre-inches*
7.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

G2 - 10/31/2022: Wheat, silage, soft dough

Field name: G2 Crop: Wheat, silage, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.95	14.36	190.74	1,236.19
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.95	14.36	190.74	1,236.19
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	169.68	34.94	229.82	980.85
Nutrient balance	63.27	-20.59	-39.08	255.34
Applied to removed ratio	1.37	0.41	0.83	1.26

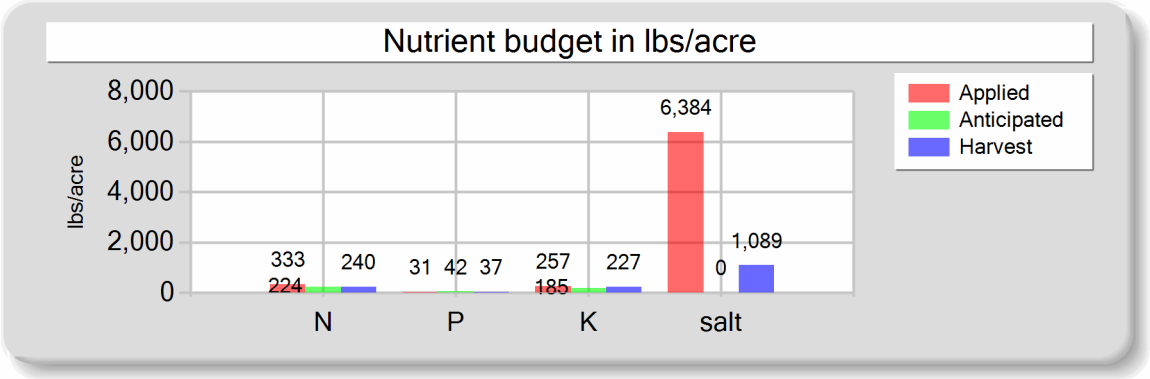
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,386,000.00	*gallons*
51.04	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

G2 - 06/25/2023: Corn, silage

Field name: G2 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.57	31.35	256.85	3,046.80
Fresh water	113.40	0.00	0.00	3,336.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	332.97	31.35	256.85	6,383.65
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	240.44	37.12	226.59	1,088.94
Nutrient balance	92.53	-5.76	30.27	5,294.71
Applied to removed ratio	1.38	0.84	1.13	5.86

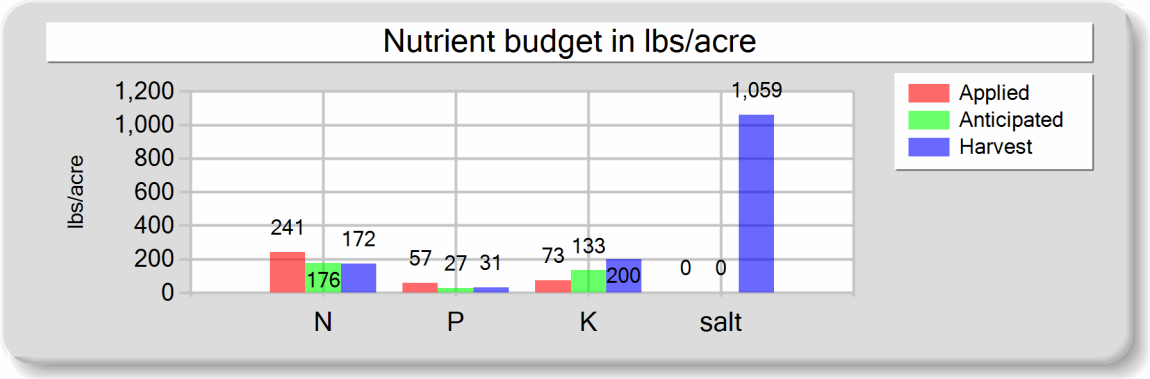
Fresh water applied	
22,248,000.00	*gallons*
819.32	*acre-inches*
22.14	*inches/acre*

Process wastewater applied	
7,854,000.00	*gallons*
289.24	*acre-inches*
7.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

Miles 1 - 11/02/2022: Wheat, silage, soft dough

Field name: Miles 1 Crop: Wheat, silage, soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.89	57.35	73.05	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.89	57.35	73.05	0.00
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	172.43	31.10	199.73	1,058.92
Nutrient balance	68.46	26.25	-126.68	-1,058.92
Applied to removed ratio	1.40	1.84	0.37	0.00

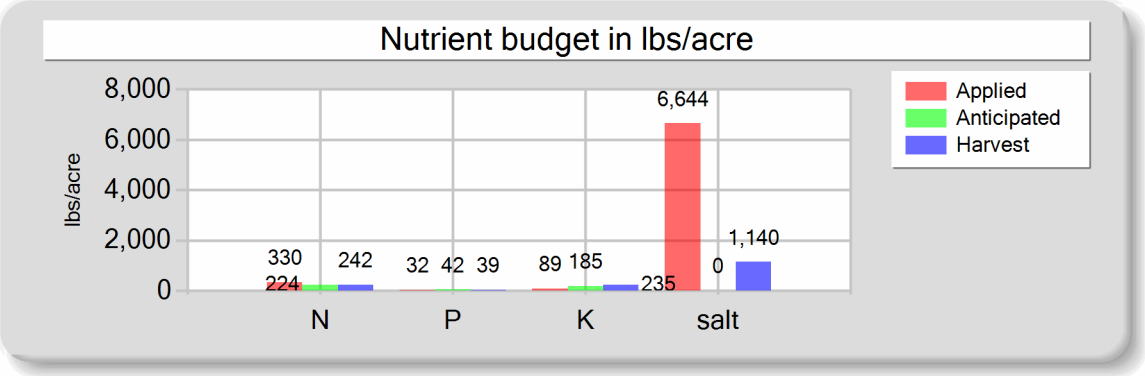
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Miles 1 - 06/27/2023: Corn, silage

Field name: Miles 1 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	125.02	25.54	34.14	1,205.87
Process wastewater	45.56	6.72	55.06	653.08
Fresh water	151.92	0.00	0.00	4,785.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.51	32.26	89.19	6,644.22
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	242.00	38.50	234.85	1,139.82
Nutrient balance	87.51	-6.24	-145.66	5,504.40
Applied to removed ratio	1.36	0.84	0.38	5.83

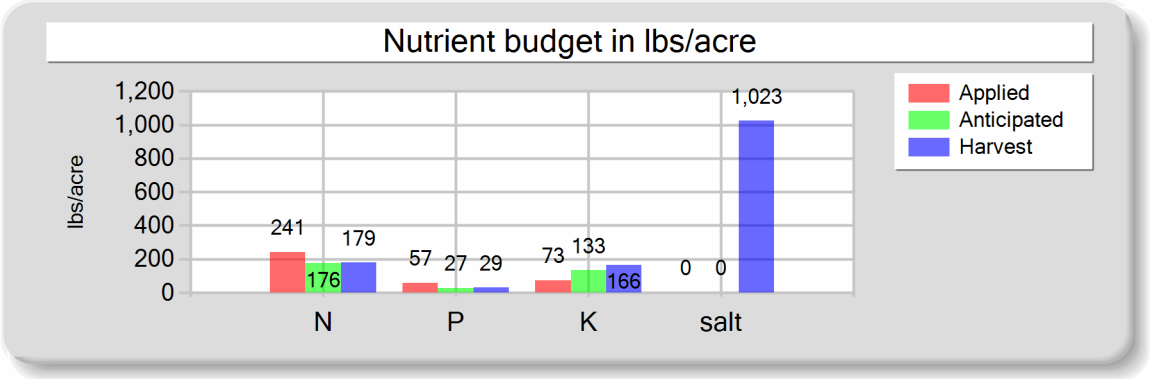
Fresh water applied	
28,872,000.00	*gallons*
1,063.26	*acre-inches*
29.53	*inches/acre*

Process wastewater applied	
1,638,000.00	*gallons*
60.32	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

Miles 2 - 11/02/2022: Wheat, silage, soft dough

Field name: Miles 2 Crop: Wheat, silage, soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.89	57.35	73.05	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.89	57.35	73.05	0.00
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	178.70	28.92	165.71	1,023.48
Nutrient balance	62.19	28.43	-92.66	-1,023.48
Applied to removed ratio	1.35	1.98	0.44	0.00

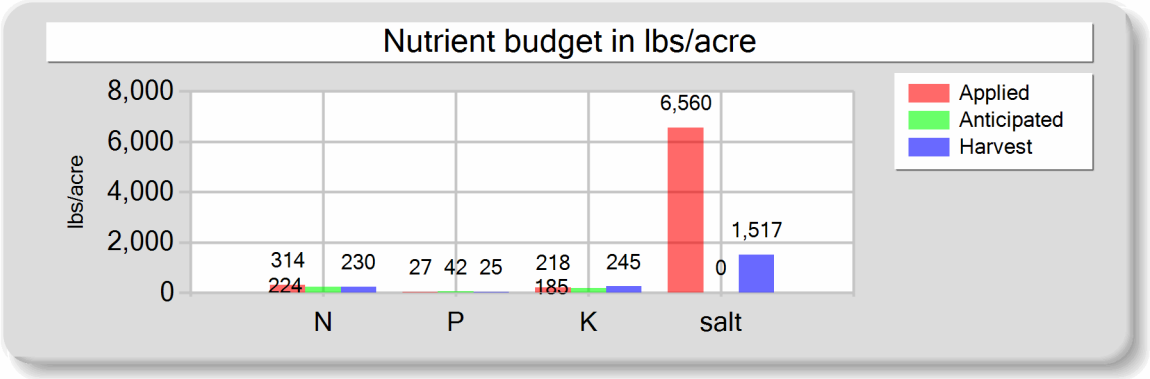
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Miles 2 - 06/25/2023: Corn, silage

Field name: Miles 2 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	180.25	26.59	217.80	2,583.61
Fresh water	126.26	0.00	0.00	3,976.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.51	26.59	217.80	6,560.49
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	229.98	25.10	244.58	1,516.96
Nutrient balance	83.53	1.49	-26.77	5,043.53
Applied to removed ratio	1.36	1.06	0.89	4.32

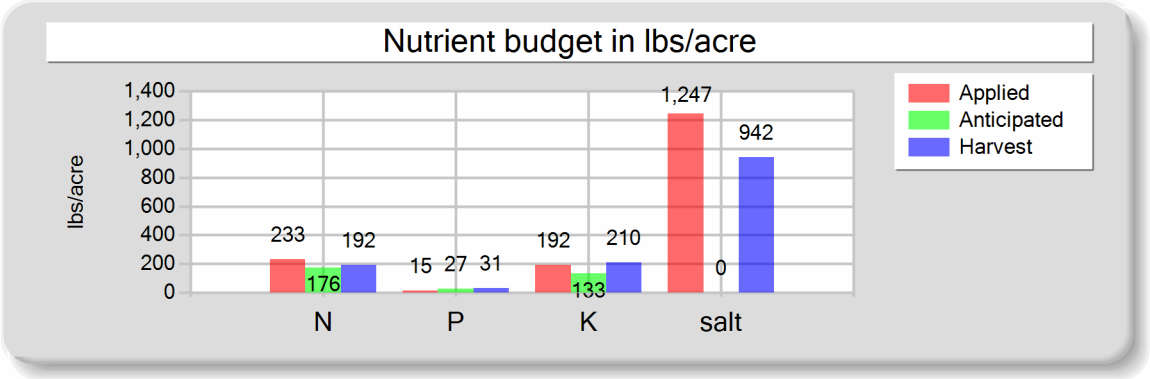
Fresh water applied	
23,328,000.00	*gallons*
859.09	*acre-inches*
24.55	*inches/acre*

Process wastewater applied	
6,300,000.00	*gallons*
232.01	*acre-inches*
6.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

Miles 3 - 11/01/2022: Wheat, silage, soft dough

Field name: Miles 3 Crop: Wheat, silage, soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.30	14.53	192.14	1,246.78
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.30	14.53	192.14	1,246.78
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	192.34	31.24	210.43	941.79
Nutrient balance	40.95	-16.71	-18.29	304.98
Applied to removed ratio	1.21	0.47	0.91	1.32

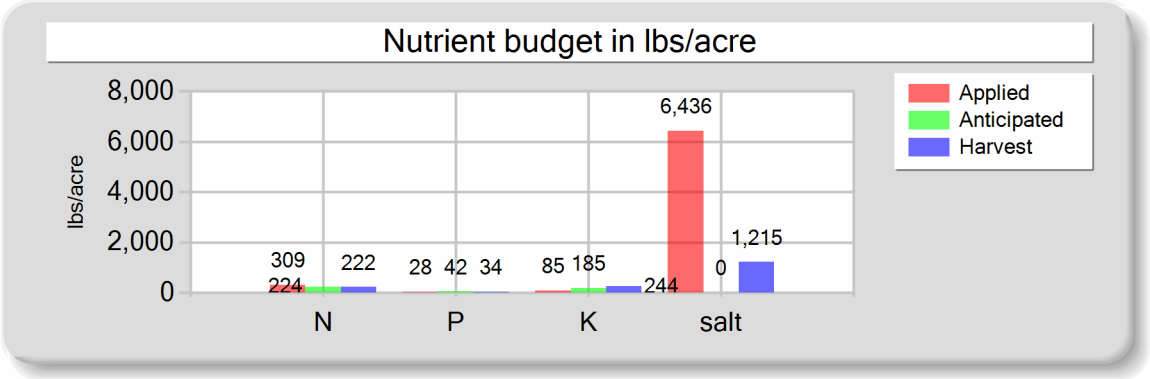
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,184,000.00 *gallons*
80.43 *acre-inches*
1.39 *inches/acre*

Total harvests for the crop
1 *harvests*

Miles 3 - 06/19/2023: Corn, silage

Field name: Miles 3 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.46	21.13	28.25	997.96
Process wastewater	47.13	6.95	56.95	675.60
Fresh water	151.21	0.00	0.00	4,762.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.81	28.09	85.21	6,436.20
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	221.71	33.97	244.36	1,214.89
Nutrient balance	87.09	-5.89	-159.15	5,221.31
Applied to removed ratio	1.39	0.83	0.35	5.30

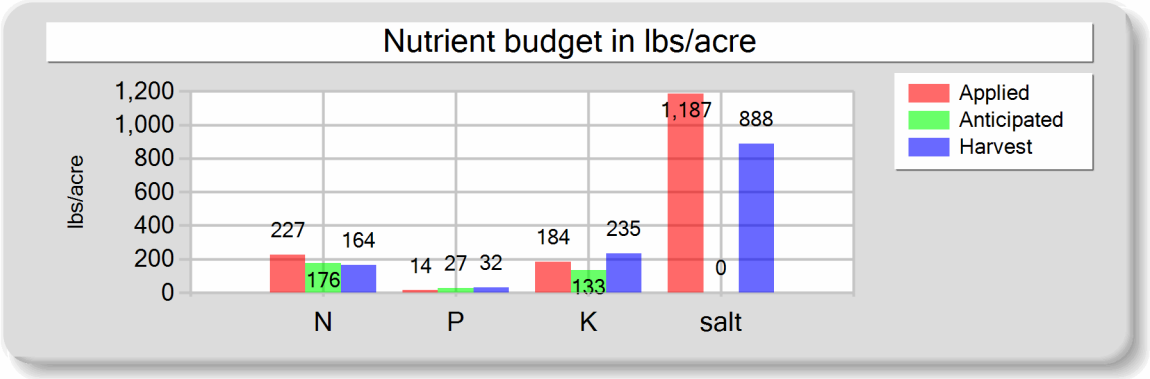
Fresh water applied	
46,296,000.00	*gallons*
1,704.92	*acre-inches*
29.40	*inches/acre*

Process wastewater applied	
2,730,000.00	*gallons*
100.54	*acre-inches*
1.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

Miles 4 - 11/01/2022: Wheat, silage, soft dough

Field name: Miles 4 Crop: Wheat, silage, soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.07	13.68	183.56	1,186.59
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.07	13.68	183.56	1,186.59
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	163.66	32.16	234.67	887.99
Nutrient balance	63.41	-18.48	-51.11	298.60
Applied to removed ratio	1.39	0.43	0.78	1.34

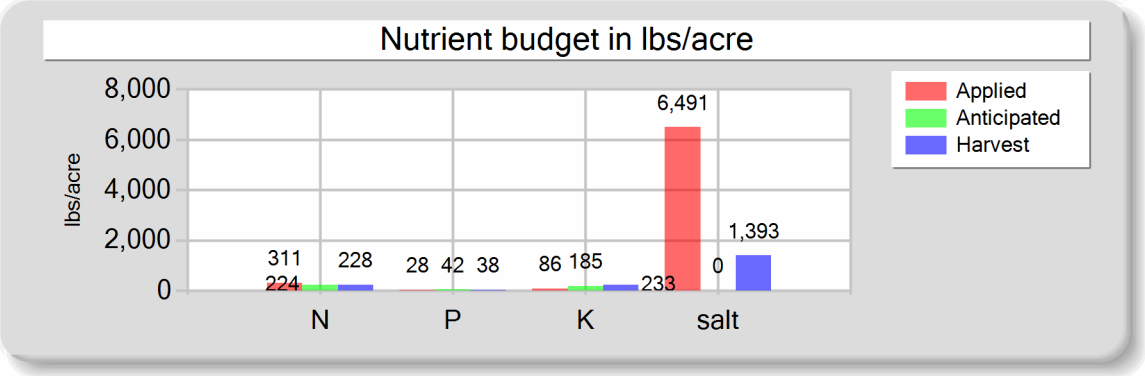
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,100,000.00	*gallons*
77.34	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Miles 4 - 06/22/2023: Corn, silage

Field name: Miles 4 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.46	21.13	28.25	997.96
Process wastewater	47.86	7.06	57.83	685.99
Fresh water	152.62	0.00	0.00	4,807.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.94	28.19	86.08	6,491.03
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	227.94	37.70	232.58	1,393.39
Nutrient balance	83.00	-9.51	-146.50	5,097.64
Applied to removed ratio	1.36	0.75	0.37	4.66

Fresh water applied	
46,728,000.00	*gallons*
1,720.83	*acre-inches*
29.67	*inches/acre*

Process wastewater applied	
2,772,000.00	*gallons*
102.08	*acre-inches*
1.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure

Sample and source description: Separator Manure

Sample date: 09/09/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,580.00	4,560.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 17.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	6,660.00	17,200.00	0.00	0.00	0.00	0.00	0.00		38.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Manure

Sample and source description: Separator Manure

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	2,880.00	3,850.00	0.00	0.00	0.00	0.00	0.00		13.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 27.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	7,340.00	24,400.00	14,400.00	13,800.00	3,210.00	5,010.00	1,000.00		33.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	2,250.00	2,200.00	0.00	0.00	0.00	0.00	0.00		8.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 03/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	515.00	339.00	0.00	0.00	70.20	710.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,020.00	4,980
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	896.00	358.00	0.00	0.00	25.70	527.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,390.00	3,100
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	120.00	81.20	0.00	0.00	17.70	145.00	89.50	131.00	78.30	1,120.00	0.00	137.00	118.00	2,570.00	1,720
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	204.00	143.00	0.00	0.00	46.30	180.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3,300.00	2,920
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

IW G

IW G

IW G

Sample description: IW G

Sample date: 07/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	22.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,260.00	665
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

IW M

IW M

Sample description: IW M

Sample date: 07/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	22.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,430.00	715
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

G1 - 10/30/2022: Wheat, silage, soft dough

G1 - 10/30/2022: Wheat, silage, soft dough

Foorage

Sample and source description: Foorage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	870.00	5,810.00		8.80
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,140.00	830.00	4,370.00		9.00
DL	100.00	100.00	100.00		0.01

G1 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,120.00	840.00	4,940.00		7.00
DL	100.00	100.00	100.00		0.01

G1 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,350.00	840.00	4,660.00		7.70
DL	100.00	100.00	100.00		0.01

G2 - 10/31/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,050.00	1,040.00	6,840.00		8.90
DL	100.00	100.00	100.00		0.01

G2 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,340.00	670.00	4,090.00		6.30
DL	100.00	100.00	100.00		0.01

Miles 1 - 11/02/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,990.00	900.00	5,780.00		9.40
DL	100.00	100.00	100.00		0.01

Miles 1 - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	700.00	4,270.00		6.60
DL	100.00	100.00	100.00		0.01

Miles 2 - 11/02/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	890.00	5,100.00		9.00
DL	100.00	100.00	100.00		0.01

Miles 2 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,940.00	430.00	4,190.00		8.90
DL	100.00	100.00	100.00		0.01

Miles 3 - 11/01/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,320.00	1,130.00	7,620.00		8.20
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,850.00	950.00	6,400.00		9.30
DL	100.00	100.00	100.00		0.01

Miles 3 - 06/19/2023: Corn, silage

Miles 3 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,720.00	570.00	4,100.00		7.28
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,430.00	650.00	3,870.00		7.10
DL	100.00	100.00	100.00		0.01

Miles 4 - 11/01/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage
Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,140.00	1,010.00	7,370.00		8.30
DL	100.00	100.00	100.00		0.01

Miles 4 - 11/01/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,520.00	1,010.00	6,900.00		8.90
DL	100.00	100.00	100.00		0.01

Miles 4 - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	650.00	4,010.00		7.70
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,530.00	670.00	3,390.00		7.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

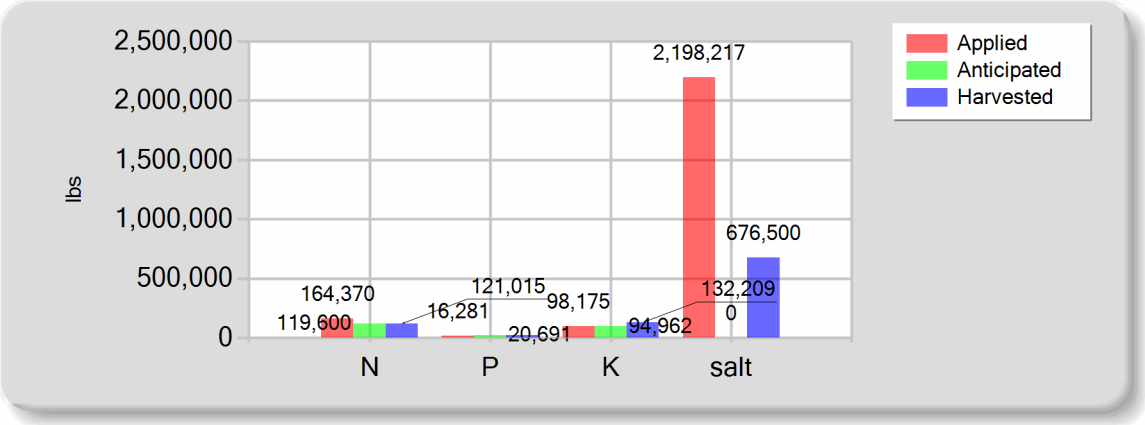
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

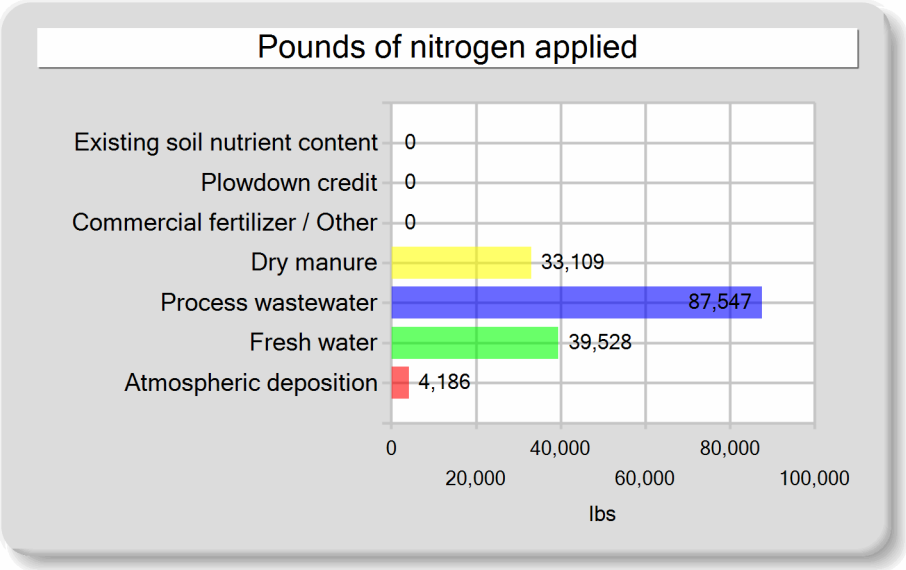
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

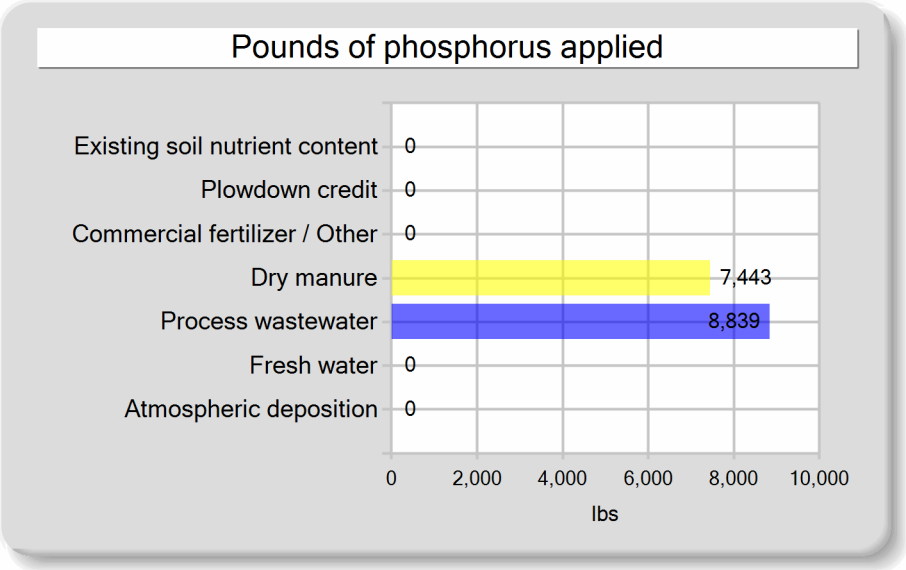
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	33,108.97	7,442.72	9,692.68	159,174.40
Process wastewater	87,546.79	8,838.69	88,482.25	818,898.86
Fresh water	39,528.43	0.00	0.00	1,220,144.14
Atmospheric deposition	4,186.00	0.00	0.00	0.00
Total nutrients applied	164,370.20	16,281.41	98,174.93	2,198,217.39
Anticipated crop nutrient removal	119,600.00	20,690.80	94,962.40	0.00
Actual crop nutrient removal	121,015.19	20,752.84	132,208.77	676,500.46
Nutrient balance	43,355.01	-4,471.43	-34,033.84	1,521,716.94
Applied to removed ratio	1.36	0.78	0.74	3.25

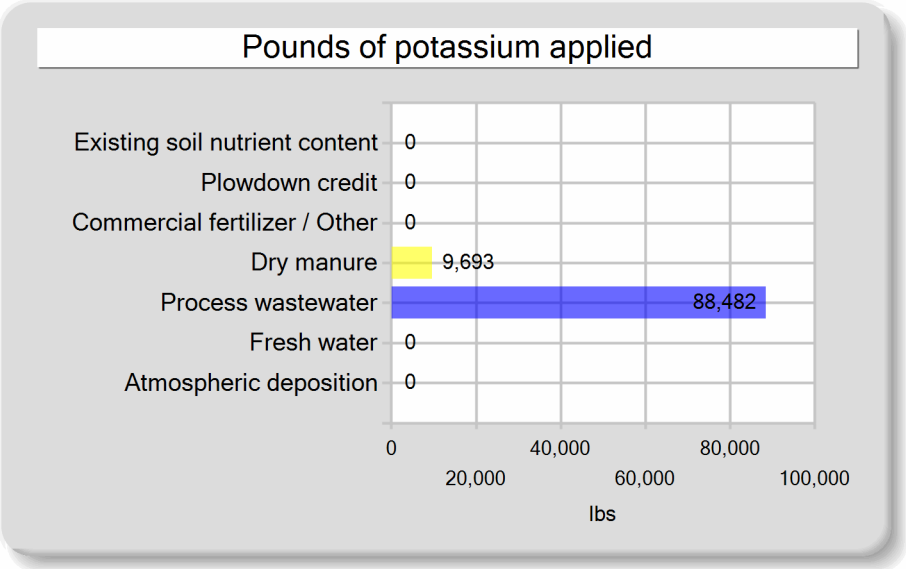
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

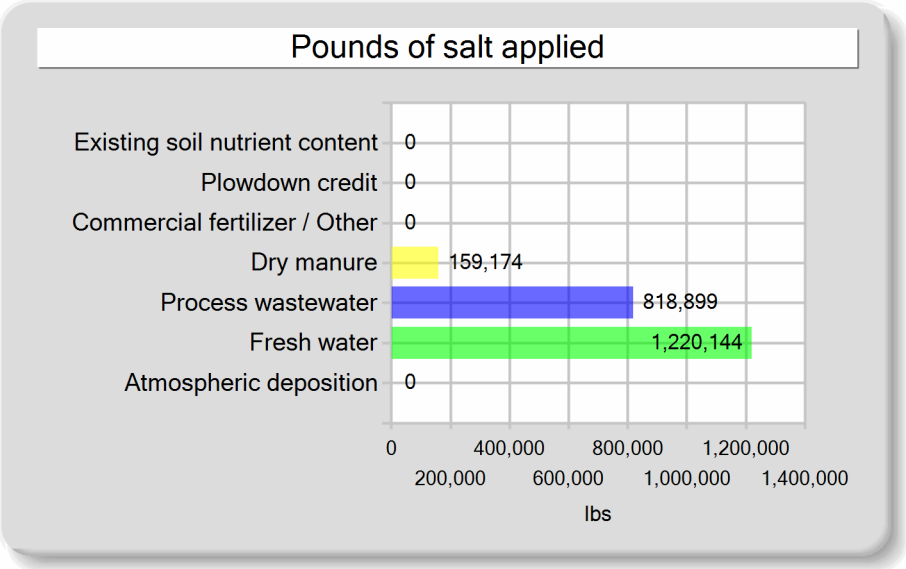


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ulysses Mendoza	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
3-25-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Ulys Dairy
8118 Sikes Rd.
Dixon, CA 95620

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 7/20/23 3:45 PM
SUBMITTED BY: Vince
ANALYZED DATE: 7/21/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 28.7c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW M	T3201026A	Electrical Conductivity (EC)	EPA 120.1 and	1.43	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.9	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	715	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	22.7	mg/L
		Soluble Salts (SALT-SOL)	Calculation	914	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	92.6			0.000 (T3201026A)	0.000 (null)	
Electrical Conductivity (EC)	94.7			2.57 (T3201025A)	2.56 (Dup)	0.117
Nitrate Nitrogen (NO3-N)	91.3	95.8 (null)	95.5 (null)			0.314
Total Dissolved Solids (TDS)	97.0			0.000 (T3202009A)	0.000 (Dup)	
Total Nitrogen (TN)	103			5.60 (T3201007A)	5.60 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Ulys Dairy
8118 Sikes Rd.
Dixon, CA 95620

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 7/20/23 3:45 PM
SUBMITTED BY: Vince
ANALYZED DATE: 7/21/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 27.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW G	T3201026B	Electrical Conductivity (EC)	EPA 120.1 and	1.26	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.8	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	665	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	22.6	mg/L
		Soluble Salts (SALT-SOL)	Calculation	808	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	92.6			0.000 (T3201026A)	0.000 (null)	
Electrical Conductivity (EC)	94.7			2.57 (T3201025A)	2.56 (Dup)	0.117
Nitrate Nitrogen (NO3-N)	91.3	95.8 (null)	95.5 (null)			0.314
Total Dissolved Solids (TDS)	97.0			0.000 (T3202009A)	0.000 (Dup)	
Total Nitrogen (TN)	103			5.60 (T3201007A)	5.60 (null)	0.000

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Ulus Dairy
Mailing address: 8118 Sikes Rd.
(not facility address) Dixon, CA
Phone#:

Authorized Copy Released To: F+R Ag

Sampled By: Vince Furtado
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	7-20-23	11:20	IW M		28.7°C						H7
2	7-20-23	11:10	IW G		27.0°						H7
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Vince Furtado	F+R Ag		7-20-23 3:45		
2nd		DENELE ANALYTICAL, INC.	7-20-23 3:45			T3201026

Required to be submitted to any agency? (YES / NO)

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Compliance Analysis Report

Customer
Ulys Dairy
8118 Sikes Rd.
Dixon, CA 95620

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/23/23 12:00 PM
SUBMITTED BY: Vince
ANALYZED DATE: 10/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy Domestic Well	T3296049A	Electrical Conductivity (EC)	EPA 120.1 and	0.578	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.959	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	370	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.4	99.4 (T3296049A)	98.2 (T3296049A)	1.21
Electrical Conductivity (EC)	96.4			0.904
Nitrate Nitrogen (NO3-N)	97.0	98.7 (T3296043A)	100 (T3296043A)	1.53

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TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Ulys Dairy
Mailing address: 8118 Sikes Road
(not facility address) Dixon, CA 95620
Phone#:

Authorized Copy Released To: F+R Ag

Sampled By: Vince Furtado
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10-23-23	8:00	Dairy Domestic Well		13.6°c		✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Vince Furtado	F+R Ag		10-23-23		
2nd		DENELE ANALYTICAL, INC.	10-23-23 12:00			T32960049

Required to be submitted to any agency? (YES / NO)